

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025**MACHINE LEARNING**

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours**Max. Marks: 60****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A**(10 Marks)**

1. a) Define overfitting in the context of machine learning. [1M] L1
- b) How does data science relate to AI and ML. [1M] L2
- c) Calculate the mean, median and mode for the given list of data [30, 35, 40, 45, 50, 55, 60, 200]. [1M] L3
- d) What is Feature Engineering? [1M] L2
- e) What is the measure of linear relationship? [1M] L2
- f) Define Mean Absolute Error (MAE). [1M] L1
- g) Define precision in the context of classification. [1M] L1
- h) What is Euclidean distance? Give example. [1M] L2
- i) How can clustering help in semi-supervised learning? [1M] L2
- j) What are principal components in PCA? [1M] L2

PART- B**(5 X 10 = 50 Marks)**

2. a) Explain supervised machine learning algorithm with suitable example. [5M] L2
 - b) Apply the concept of reinforcement learning to a real-world scenario and describe its components. [5M] L3
- OR**
3. a) Explain Unsupervised Machine Learning algorithm in detail with an example. [5M] L2
 - b) Discuss various applications of machine learning. [5M] L2

4. A company monthly sales revenue (in thousands of dollars) for the past 12 months is given below:

12, 15, 14, 16, 17, 20, 25, 22, 18, 19, 100, 210.

- i) Calculate the mean and median of the sales revenue.
- ii) Identify whether the distribution of sales revenue is skewed.
- iii) Based on the skewness, explain which measure (mean or median) would better represent the typical monthly sales revenue. [10M] L3

OR

5. a) Given the following data set, calculate the correlation between the columns Age and Income. What does the correlation value indicate? [5M] L3

Name	Age	Income (\$)
John	25	40000
Sarah	30	50000
Mike	35	55000
Emma	40	60000
Liam	28	45000

- b) Explain How a box plot helps in identifying outliers using the interquartile range (IQR). Given the data: 22, 26, 17, 19, 34, 64, 23, 17, 21, 35. Identify the outliers and sketch the box plot. [5M] L3

6. a) Explain Regression with stats models. [5M] L2

- b) Define and calculate MAE, MSE, and RMSE for the give data set: [5M] L3

Actual Salary	20	30	45	25	70	65	68	50	55	35
Predicted Salary	25	32	40	30	67	60	65	49	50	32

OR

7. a) Evaluate the impact of overfitting and underfitting on model performance, and propose strategies to address each issue. [5M] L3

- b) Explain the concept of linear relationship between variables. How is it measured in regression analysis. [5M] L2

8. a) Describe logistic regression in detail. Explain the concept of log-odds and how the sigmoid function transforms outputs into probabilities. [5M] L2

- b) What is a classification problem in machine learning? Differentiate between binary and multi-class classification with suitable examples. [5M] L2

OR

9. a) Discuss the concept of Gradient Descent. How is it used to optimize classification algorithms like Logistic Regression? Provide a basic outline or equation. [5M] L3

- b) Explain the concept of Support Vector Machines (SVM). [5M] L2

10. a) Explain the K-Means clustering algorithm step-by-step. How does the algorithm iteratively refine the cluster centroids until convergence? [7M] L3

- b) What are the limitations of K-Means algorithm? [3M] L2

OR

11. a) Identify the core points, border points, and noise points and form clusters for the following data points using DBSCAN algorithm. P1(1,2), P2(2,2), P3(2,3), P4(8,7), P5(8,8), P6(25,80) $\epsilon \leq 1.41$, Min Pts = 3. [7M] L3

- b) What is the curse of dimensionality? [3M] L2

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B.Tech II Year II Semester Regular Examinations, June/July-2025

MACHINE LEARNING (INFORMATION TECHNOLOGY)

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) What is the difference between Machine Learning and Deep Learning? [1M] L2
- b) Give two real life problems that can be solved using machine learning. [1M] L2
- c) Calculate Mean Absolute deviation for the data: 7, 8, 4, 10, 12, 18. [1M] L2
- d) Define Outliers. [1M] L2
- e) What is Regularization? [1M] L2
- f) Explain how overfitting affects the performance of a regression model. [1M] L2
- g) Define accuracy in classification. [1M] L2
- h) What is the formula for Information Gain? [1M] L2
- i) What is clustering in unsupervised learning? [1M] L2
- j) Name two approaches to dimensionality reduction. [1M] L2

PART- B

(5 X 10 = 50 Marks)

2. a) Discuss essential python libraries used for machine learning. [5M] L2
- b) What are the different types of machine learning algorithms? Explain. [5M] L2

OR

3. a) What are the main challenges of Machine Learning? Explain. [5M] L2
- b) A layman can segregate different types of fruits but he cannot define names or labels for the fruits.
What type of machine learning algorithm is used discuss in detail. [5M] L2
4. a) Discuss various data visualization tools and techniques used in EDA. [5M] L2
- b) Write Python code for Plotting Box and Whisker plot. [5M] L3

OR

5. a) Using the provided dataset on individuals, perform an exploratory data analysis to identify trends and patterns. [6M] L3

ID	Age	Gender	Income	City	Employment Status
1	25	Male	55000	New York	Employed
2	NaN	Female	62000	Los Angeles	Employed
3	30	NaN	NaN	Chicago	Unemployed
4	40	Female	70000	NaN	Employed
5	35	Male	58000	San Francisco	NaN

- i) Calculate summary statistics for numerical features.
- ii) Identify and handle missing values and categorical variables.
- iii) Discuss any potential correlations between Age and Income. What conclusions can you make based on your analysis.

b) How do you apply one-hot encoding to a categorical variable? Explain with an example. [4M] L2

6. A biologist assumes that there is linear relationship between the amount of fertilizer supplied to the plants and subsequent yield of crop obtained. Eight plants of same variety were selected at random and treated with X amount of fertilizer and Y amount of crop yields are recorded as given below. [10M]L3

Plant	A	B	C	D	E	F	G	H
X	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Y	3.9	4.4	5.8	6.6	7.0	7.1	7.3	7.7

- a) Calculate the equation of the regression line of Y on X.
- b) Estimate the yield of plant treated with 3.2 grams of fertilizer.

OR

7. a) Compare and contrast Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression. In which scenarios would each method be most appropriate? [5M] L4

b) What are regression coefficients? How are they determined, and what is the significance of their values in a regression model? Explain. [5M] L2

8. Explain the different evaluation metrics for classification models. For the following confusion matrix, calculate Accuracy, Precision, Recall and F1 score. [10M] L3

		Predicted	
		0	1
Actual	0	30	12
	1	8	56

OR

9. a) Explain the concept of the receiver operating characteristic (ROC) curve and its importance in evaluating classification models. How does the area under the ROC curve (AUC) reflect model performance. [5M] L3

b) Using KNN ($K=3$), find the quality of paper as good or bad with $x=3$ and $y=7$ from the given dataset:

X	7	7	3	1
Y	7	4	4	4
Quality	Bad	Bad	Good	Good

10. a) Discuss the limitations of K-Means clustering. How can these limitations affect the performance of a clustering model? Suggest ways to overcome them. [5M] L2

b) What are the algorithmic steps of DBSCAN, and how does it detect clusters in data? [5M] L2

OR

11. How does PCA help in reducing dimensionality while preserving most of the information in a dataset? Explain with the concept of variance and principal components. [10M] L3

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Kmit

Code No: 235AH

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE Affiliated to JNTU-H, Hyderabad-500 029

B.Tech III Year I Semester Regular Examinations, December-2025

MACHINE LEARNING

(CSE (DATA SCIENCE))

Time: 3 hours

Max. Marks: 60.

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(10×1 = 10 Marks)

1. a) Define Machine Learning. [1M]L1
- b) What is supervised learning? [1M]L1
- c) Define hypothesis space. [1M]L1
- d) What is overfitting? [1M]L2
- e) Define activation function. [1M]L1
- f) What is Hidden Layer? [1M]L1
- g) Define clustering. [1M]L1
- h) What is CNN algorithm? [1M]L1
- i) Define GRU. [1M]L1
- j) What is RNN-learning? [1M]L1

PART- B

(5 X 10 =50 Marks)

2. a) Explain the different types of Machine Learning with examples. [5M]L2
- b) Describe the applications of Machine Learning. [5M]L3

OR

3. a) Explain the concept of training, validation, and testing data. [5M]L2
 - b) Explain linear separable problem in neural network with XOR example. [5M]L2
4. a) Discuss gradient descent algorithm. [5M]L3
- b) What's the role of Weights and Bias in an NN [5M]L4

OR

5. a) Explain logistic regression classification in neural network, activation functions used and its applications. [5M]L2
- b) Why do we flatten the Image in Neural Network? Explain with example of MNIST dataset. [5M] L4

6. a) What is an overfitting, how does Dropout will reduce the overfitting problem illustrate with an example. [5M]L2
b) Describe the function of each component in a basic CNN model:
Conv2D → ReLU → MaxPooling → Flatten → Dense. [5M]L3
OR
7. a) Explain Naïve Bayes classifier with example. [5M]L2
b) Discuss advantages and disadvantages of Naïve Bayes. [5M]L4
8. a) Discuss the LeNet-5 architecture layer by layer. [5M]L2
b) Summarize the key architectural differences between LeNet-5, AlexNet, and VGGNet. [5M]L2
OR
9. a) Explain hierarchical clustering techniques. [5M]L2
b) Compare VGG16 and RESTNET [5M]L4
10. a) Explain LSTM learning. [5M]L2
b) Describe Markov Decision Process. [5M]L3
OR
11. a) Explain why GRU sentiment classification with suitable example [5M]L3
b) Discuss applications RNN learning applications. [5M]L3

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